

APP SERVER AND WEB SERVER

1) Apache HTTPD

```
sudo yum install httpd -y
sudo systemctl start httpd
sudo systemctl status httpd
sudo systemctl enable httpd
```

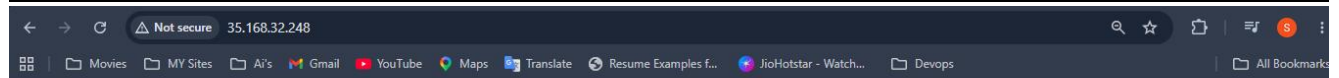
2) TO COPY THE CODE httpd

/var/www/html/

→ to assign port number {81}
- - /etc/httpd/conf/httpd.conf

```
[ec2-user@ip-172-31-4-184 apache-tomcat-9.0.121]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-08-19 03:55:38 UTC; 26min ago
     Docs: man:httpd.service(8)
   Main PID: 1742 (httpd)
   Status: "Total requests: 42; Idle/Busy workers 100/0; Requests/sec: 0.0271; Bytes served/sec: 1.2KB/sec"
    Tasks: 177 (limit: 1059)
   Memory: 22.2M
      CPU: 1.639s
   CGroup: /system.slice/httpd.service
           └─1742 /usr/sbin/httpd -DFOREGROUND
             └─1743 /usr/sbin/httpd -DFOREGROUND
               └─1744 /usr/sbin/httpd -DFOREGROUND
                 └─1749 /usr/sbin/httpd -DFOREGROUND
                   └─1771 /usr/sbin/httpd -DFOREGROUND

Aug 19 03:55:38 ip-172-31-4-184.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Aug 19 03:55:38 ip-172-31-4-184.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Aug 19 03:55:38 ip-172-31-4-184.ec2.internal httpd[1742]: Server configured, listening on: port 81
[ec2-user@ip-172-31-4-184 apache-tomcat-9.0.121]$ cd /var/www/html/
[ec2-user@ip-172-31-4-184 html]$ ls
blog.html  checkout.html  contact.html  css  ecomm  fonts  img  index.html  js  regular-page.html  scss  shop.html  single-blog.html  single-product-details.html  style.css
[ec2-user@ip-172-31-4-184 html]$
```



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-50%

MANGO

PETITE Crepe Wrap Mini Dress

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35.168.32.248/#

↑

APP SERVER AND WEB SERVER

3) Nginx

```
sudo yum -y install nginx
sudo systemctl start nginx
sudo systemctl status nginx
sudo systemctl enable nginx
```

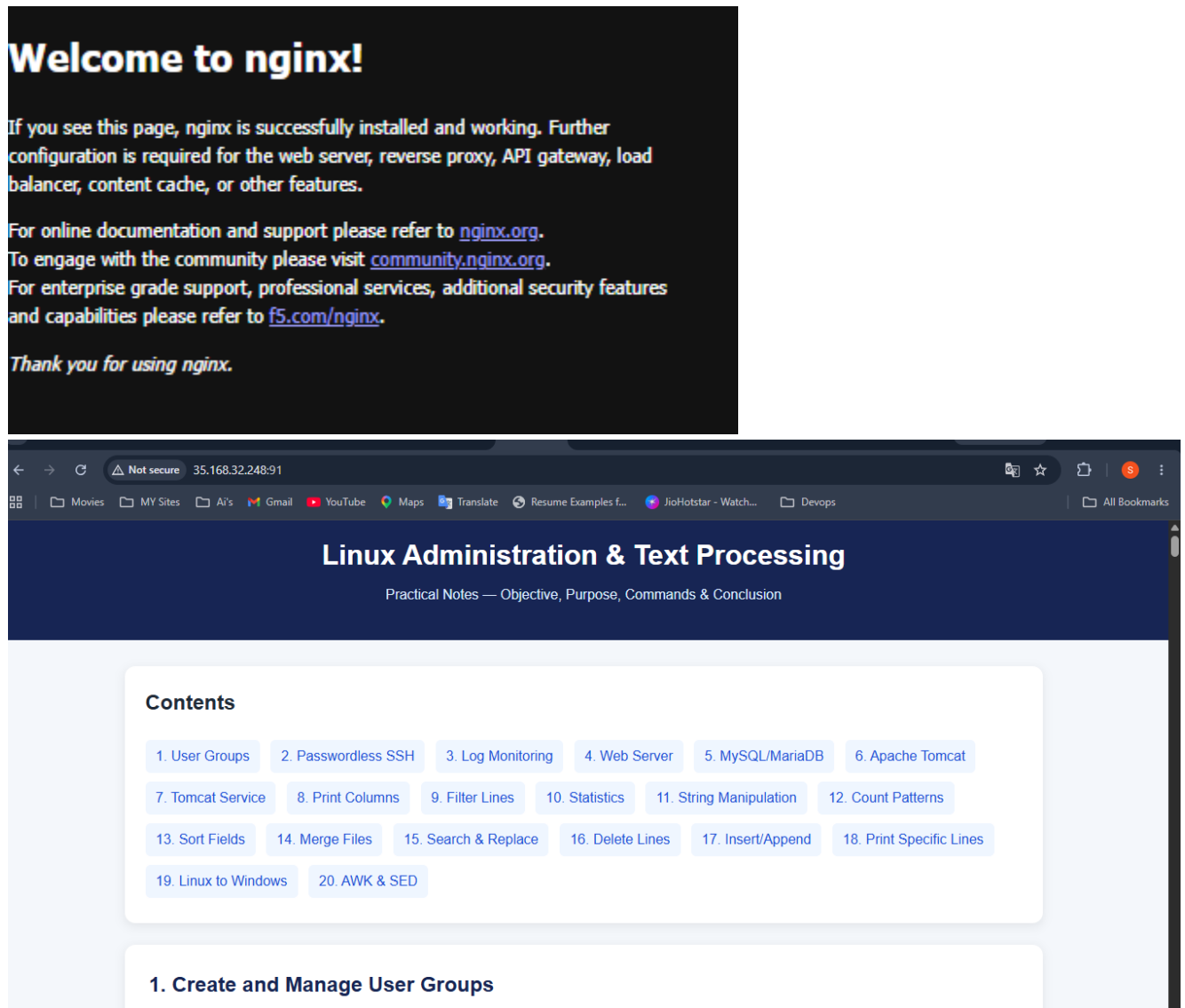
4) TO COPY THE CODE TO nginx

```
/usr/share/nginx/html/
→ to assign port number {91}
- - /etc/nginx/nginx.conf
```

```
[ec2-user@ip-172-31-4-184 ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-08-19 03:59:39 UTC; 27min ago
     Main PID: 2390 (nginx)
        Tasks: 3 (limit: 1059)
       Memory: 5.3M
          CPU: 79ms
      CGroup: /system.slice/nginx.service
              └─2390 "nginx: master process /usr/sbin/nginx"
                └─2391 "nginx: worker process"
                  └─2392 "nginx: worker process"

Aug 19 03:59:39 ip-172-31-4-184.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Aug 19 03:59:39 ip-172-31-4-184.ec2.internal nginx[2355]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Aug 19 03:59:39 ip-172-31-4-184.ec2.internal nginx[2355]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Aug 19 03:59:39 ip-172-31-4-184.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[ec2-user@ip-172-31-4-184 ~]$ cd /usr/share/nginx/html/
[ec2-user@ip-172-31-4-184 html]$ ls
index.html
[ec2-user@ip-172-31-4-184 html]$ cat index.html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Linux Administration & Text Processing - Practical Notes</title>
<style>
  * { box-sizing: border-box; }
  body {
    margin: 0;
    font-family: Arial, Helvetica, sans-serif;
    background: #f4f7fb;
    color: #1f2937;
    line-height: 1.6;
  }
  header {
    background: #172554;
    color: white;
    padding: 40px 20px;
    text-align: center;
  }
  header h1 { margin: 0 0 10px; font-size: 32px; }
  header p { margin: 0; color: #dbeafe; }
  .container {
    max-width: 1100px;
    margin: 30px auto;
    padding: 0 20px 50px;
  }
}
```

APP SERVER AND WEB SERVER



5) Apache Tomcat

I'm downloading apache

→ cd /opt/ wget <https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.121/bin/apache-tomcat-9.0.121.tar.gz>

→ tar -xvf apache-tomcat-9.0.121.tar.gz

➔ Cd bin/

→ ./startup.sh

→ publicip:8080

-- To assign port number {8181}

/apache-tomcat-9.0.80/conf/server.xml

APP SERVER AND WEB SERVER

```
[root@ip-172-31-4-184 webapps]# ls
META-INF  ROOT  WEB-INF  docs  examples  hello.jsp  host-manager  images  index.html  manager  sample  sample.war
[root@ip-172-31-4-184 webapps]# cd ..
[root@ip-172-31-4-184 apache-tomcat-9.0.121]# cd .
[root@ip-172-31-4-184 apache-tomcat-9.0.121]# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
[root@ip-172-31-4-184 apache-tomcat-9.0.121]# cd ..
[root@ip-172-31-4-184 opt]# ls
apache-tomcat-9.0.121  apache-tomcat-9.0.121.tar.gz  aws
[root@ip-172-31-4-184 opt]# cd apache-tomcat-9.0.121/
[root@ip-172-31-4-184 apache-tomcat-9.0.121]# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
[root@ip-172-31-4-184 apache-tomcat-9.0.121]# cd bin/
[root@ip-172-31-4-184 bin]# ls
bootstrap.jar  catalina.sh  commons-daemon-native.tar.gz  configtest.sh  digest.sh  setclasspath.bat  shutdown.sh  tomcat-juli.jar  tool-wrapper.sh
catalina-tasks.xml  ciphers.bat  commons-daemon.jar  daemon.sh  makebase.bat  setclasspath.sh  startup.bat  tomcat-native.tar.gz  version.bat
catalina.bat  ciphers.sh  configtest.bat  digest.bat  makebase.sh  shutdown.bat  startup.sh  tool-wrapper.bat  version.sh
[root@ip-172-31-4-184 bin]# cd ..
[root@ip-172-31-4-184 apache-tomcat-9.0.121]# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
[root@ip-172-31-4-184 apache-tomcat-9.0.121]# cd webapps/
[root@ip-172-31-4-184 webapps]# ls
META-INF  ROOT  WEB-INF  docs  examples  hello.jsp  host-manager  images  index.html  manager  sample  sample.war
[root@ip-172-31-4-184 webapps]# cd sample/
[root@ip-172-31-4-184 sample]# ls
META-INF  WEB-INF  hello.jsp  images  index.html
[root@ip-172-31-4-184 sample]# vi index.html
[root@ip-172-31-4-184 sample]# pwd
/opt/apache-tomcat-9.0.121/webapps/sample
[root@ip-172-31-4-184 sample]#
```

← → ↻ 🔒 Not secure 35.168.32.248:8080

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Apache Tomcat/9.0.121



If you're seeing this, you've successfully installed Tomcat. Congratulations!

 Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

Server Status
Manager App
Host Manager

Developer Quick Start

Tomcat Setup	Realms & AAA	Examples	Servlet Specifications
First Web Application	JDBC DataSources		Tomcat Versions

Managing Tomcat

For security, access to the `manager` webapp is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 9.0 access to the `manager`

Documentation

- [Tomcat 9.0 Documentation](#)
- [Tomcat 9.0 Configuration](#)
- [Tomcat Wiki](#)

Getting Help

FAQ and Mailing Lists

The following mailing lists are available:

- [tomcat-announce](#)
Important announcements, releases, security



Sample "Hello, This is sunith" Application

This is the home page for a sample application used to illustrate the source directory organization of a web application Developer's Guide.

To prove that they work, you can execute either of the following links:

- To a [JSP page](#).
- To a [servlet](#).

APP SERVER AND WEB SERVER

Configured HA Proxy Server:

Launch 3 ec2 instances name as Server1, Server-2, HA-Proxy-Server.

Instances (1/3) Info Last updated 41 minutes ago Connect Instance state Actions Launch instances

Saved filter sets
Choose filter set Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elasti
☐	server -1	i-0b947fd569cb03261	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-100-61-171-41.co...	100.61.171.41	-
☐	server -2	i-0b5b752a35a0191d6	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-3-210-204-236.co...	3.210.204.236	-
☒	HA-Proxy-Server.	i-00be99f0e5fa60be7	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-44-199-190-154.co...	44.199.190.154	-

-- sudo yum -y install httpd

```
Last login: Wed Aug 19 05:34:36 2026 from 103.160.27.181
[ec2-user@ip-172-31-5-147 ~]$
[ec2-user@ip-172-31-5-147 ~]$ sudo -i
[root@ip-172-31-5-147 ~]# systemctl start https
Failed to start https.service: Unit https.service not found.
[root@ip-172-31-5-147 ~]# systemctl start httpd
[root@ip-172-31-5-147 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-08-19 05:52:03 UTC; 13s ago
     Docs: man:httpd.service(8)
  Main PID: 26759 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 1059)
    Memory: 13.8M
       CPU: 71ms
    CGroup: /system.slice/httpd.service
            └─26759 /usr/sbin/httpd -DFOREGROUND
              └─26760 /usr/sbin/httpd -DFOREGROUND
                └─26761 /usr/sbin/httpd -DFOREGROUND
                  └─26762 /usr/sbin/httpd -DFOREGROUND
                    └─26764 /usr/sbin/httpd -DFOREGROUND

Aug 19 05:52:03 ip-172-31-5-147.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Aug 19 05:52:03 ip-172-31-5-147.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Aug 19 05:52:03 ip-172-31-5-147.ec2.internal httpd[26759]: Server configured, listening on: port 80
[root@ip-172-31-5-147 ~]#
```

vi /etc/hosts

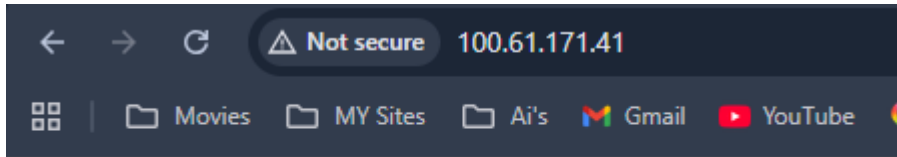
added haproxy server pub ip

```
[root@ip-172-31-5-147 ~]# cat /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost6 localhost6.localdomain6
[root@ip-172-31-5-147 ~]# vi /etc/hosts
[root@ip-172-31-5-147 ~]# cat /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost6 localhost6.localdomain6
44.199.190.154 load_balancer
[root@ip-172-31-5-147 ~]# ping load_balancer -c 4
PING load_balancer (44.199.190.154) 56(84) bytes of data.

--- load_balancer ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3149ms
```

APP SERVER AND WEB SERVER

Result:



It works!

Server-2 :

Sudo yum -y Install nginx

```
Complete!
[root@ip-172-31-12-247 ~]# amazon-linux-extras install nginx1 -y
-bash: amazon-linux-extras: command not found
[root@ip-172-31-12-247 ~]# yum amazon-linux-extras install nginx1 -y
No such command: amazon-linux-extras. Please use /usr/bin/yum --help
It could be a YUM plugin command, try: "yum install 'dnf-command(amazon-linux-extras)'"
[root@ip-172-31-12-247 ~]# cat /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost6 localhost6.localdomain6
[root@ip-172-31-12-247 ~]# vi /etc/hosts
[root@ip-172-31-12-247 ~]# ping load_balancer -c 4
PING load_balancer (44.199.190.154) 56(84) bytes of data.

--- load_balancer ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3139ms

[root@ip-172-31-12-247 ~]# systemctl start nginx
[root@ip-172-31-12-247 ~]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-08-19 06:03:16 UTC; 13s ago
     Process: 27008 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 27009 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 27010 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
    Main PID: 27011 (nginx)
      Tasks: 3 (limit: 1059)
     Memory: 3.7M
        CPU: 60ms
    CGroup: /system.slice/nginx.service
            └─27011 "nginx: master process /usr/sbin/nginx"
              └─27012 "nginx: worker process"
                └─27013 "nginx: worker process"

Aug 19 06:03:16 ip-172-31-12-247.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Aug 19 06:03:16 ip-172-31-12-247.ec2.internal nginx[27009]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Aug 19 06:03:16 ip-172-31-12-247.ec2.internal nginx[27009]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Aug 19 06:03:16 ip-172-31-12-247.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[root@ip-172-31-12-247 ~]#
```

vi /etc/hosts

added haproxy sever pub ip

APP SERVER AND WEB SERVER

```
Complete!
[root@ip-172-31-12-247 ~]# amazon-linux-extras install nginx1 -y
-bash: amazon-linux-extras: command not found
[root@ip-172-31-12-247 ~]# yum amazon-linux-extras install nginx1 -y
No such command: amazon-linux-extras. Please use /usr/bin/yum --help
It could be a YUM plugin command, try: "yum install 'dnf-command(amazon-linux-extras)'"
[root@ip-172-31-12-247 ~]# cat /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost6 localhost6.localdomain6
[root@ip-172-31-12-247 ~]# vi /etc/hosts
[root@ip-172-31-12-247 ~]# ping load_balancer -c 4
PING load_balancer (44.199.190.154) 56(84) bytes of data.

--- load_balancer ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3139ms
```

Result:

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.

To engage with the community please visit community.nginx.org.

For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

HA-Proxy-Server :

-- sudo yum install -y haproxy

-- vi /etc/hosts

→ Add Server-1, Server-2 Public IP's

```
lec2-user@ip-172-31-3-180 ~]$ sudo -i
[root@ip-172-31-3-180 ~]# yum update -y
Amazon Linux 2023 Kernel Livepatch repository                    581 kB/s | 69 kB    00:00
Last metadata expiration check: 0:00:01 ago on Wed Aug 19 06:08:37 2026.
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-3-180 ~]# yum install haproxy -y
Last metadata expiration check: 0:00:26 ago on Wed Aug 19 06:08:37 2026.
Dependencies resolved.
=====
Package                Architecture      Version           Repository         Size
-----
Installing:
haproxy                 x86_64            3.0.23-2.amzn2023.0.1  amazonlinux        2.7 M
Transaction Summary
-----
Install 1 Package

Total download size: 2.7 M
Installed size: 8.5 M
Downloading Packages:
haproxy-3.0.23-2.amzn2023.0.1.x86_64.rpm                        13 MB/s | 2.7 MB    00:00
-----
Total
Running transaction check
Transaction check succeeded.
Running transaction test
```


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vi /etc/haproxy/haproxy.cfg

Add Server-1, Server-2 public IP's

```
#-----
defaults
    mode                http
    log                 global
    option              httplog
    option              dontlognull
    option http-server-close
    option forwardfor   except 127.0.0.0/8
    option              redispatch
    retries             3
    timeout http-request 10s
    timeout queue       1m
    timeout connect     10s
    timeout client      1m
    timeout server      1m
    timeout http-keep-alive 10s
    timeout check       10s
    maxconn             3000

#-----
# main frontend which proxys to the backends
#-----
frontend main
    bind *:80
    acl url_static      path_beg       -i /static /images /javascript /stylesheets
    acl url_static      path_end       -i .jpg .gif .png .css .js

    use_backend static   if url_static
    default_backend      app

#-----
# static backend for serving up images, stylesheets and such
#-----
backend static
    balance roundrobin
    server static 127.0.0.1:4331 check

#-----
# round robin balancing between the various backends
#-----
backend app
    balance roundrobin
    server app1 127.0.0.1:5001 check
    server app2 127.0.0.1:5002 check
    server app3 127.0.0.1:5003 check
    server app4 127.0.0.1:5004 check
    server app5 100.61.171.41:80 check
    server app6 3.210.204.236:80 check
```

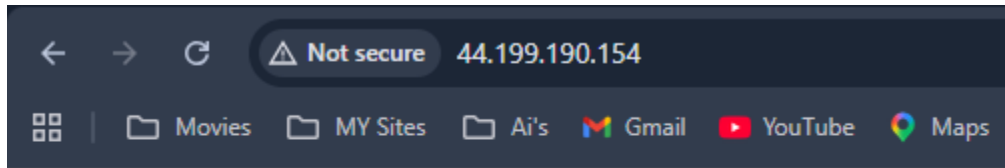
systemctl enable haproxy

systemctl start haproxy

systemctl status haproxy

APP SERVER AND WEB SERVER

result:



It works!